Cryptography

AES & Galois Fields

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Introduction

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Advanced Encryption Standard (AES)



UC 1. Cryptographic Foundations and Symmetric Cryptography

- Introduction to Cryptography and Security
- Classical Ciphers and Cryptanalysis
- Modular Arithmetic
- Shift and Affine Ciphers
- Symmetric Cryptography
- Stream Ciphers and basic attacks
- Block Ciphers and AES
- AES and Modes of Operation

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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



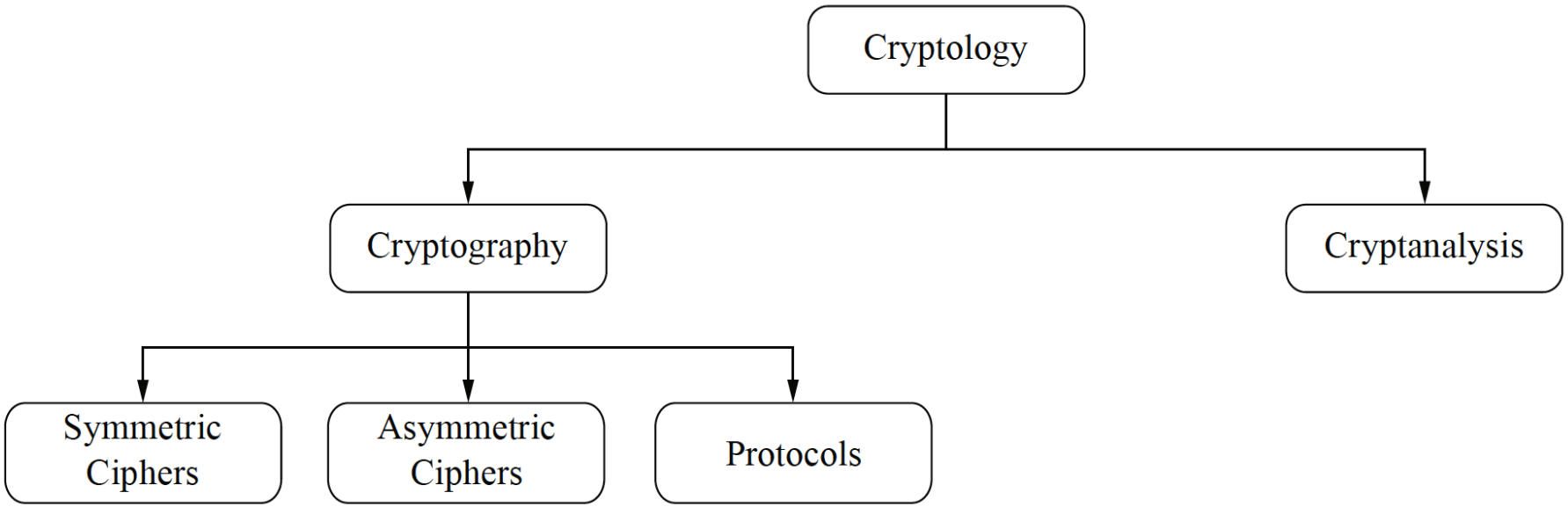
Key
interchange

Phil
Zimmerman



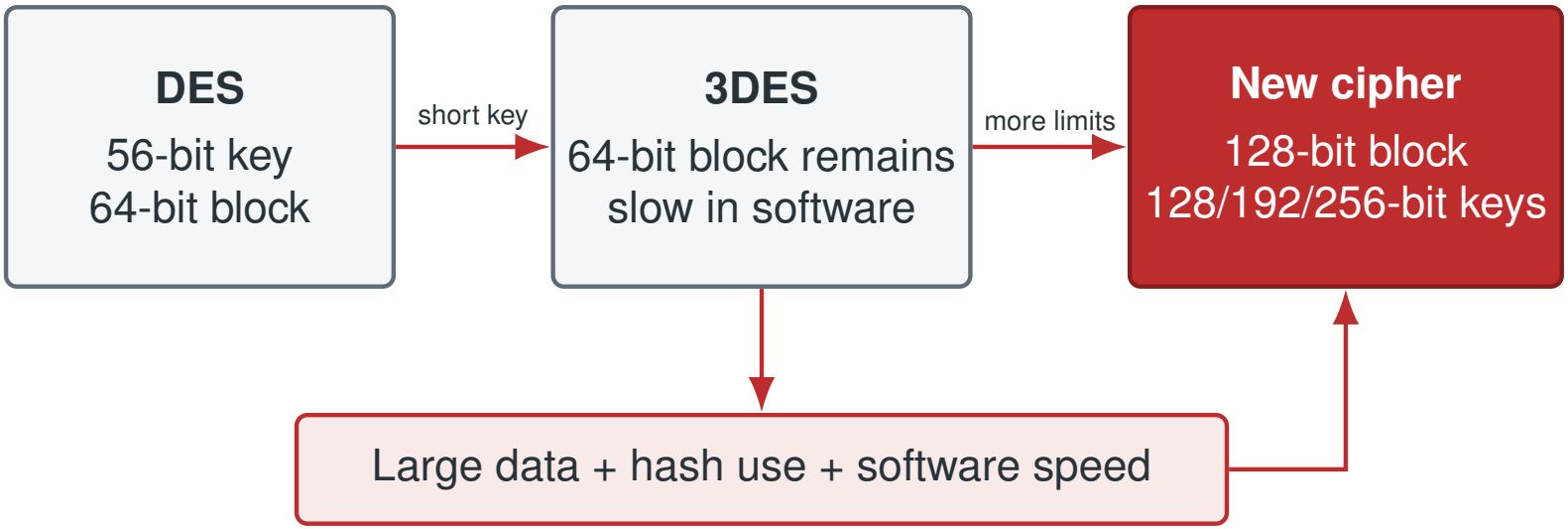
PGP
Encryption

Advanced Encryption Standard (AES)



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Why Did AES Replace DES / 3DES?



3DES improved key security, but its 64-bit block and speed were still problems.

NIST decided to design a new block cipher.

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How AES Was Selected

Required: 128-bit block; 128/192/256-bit keys

Public review by NIST + community



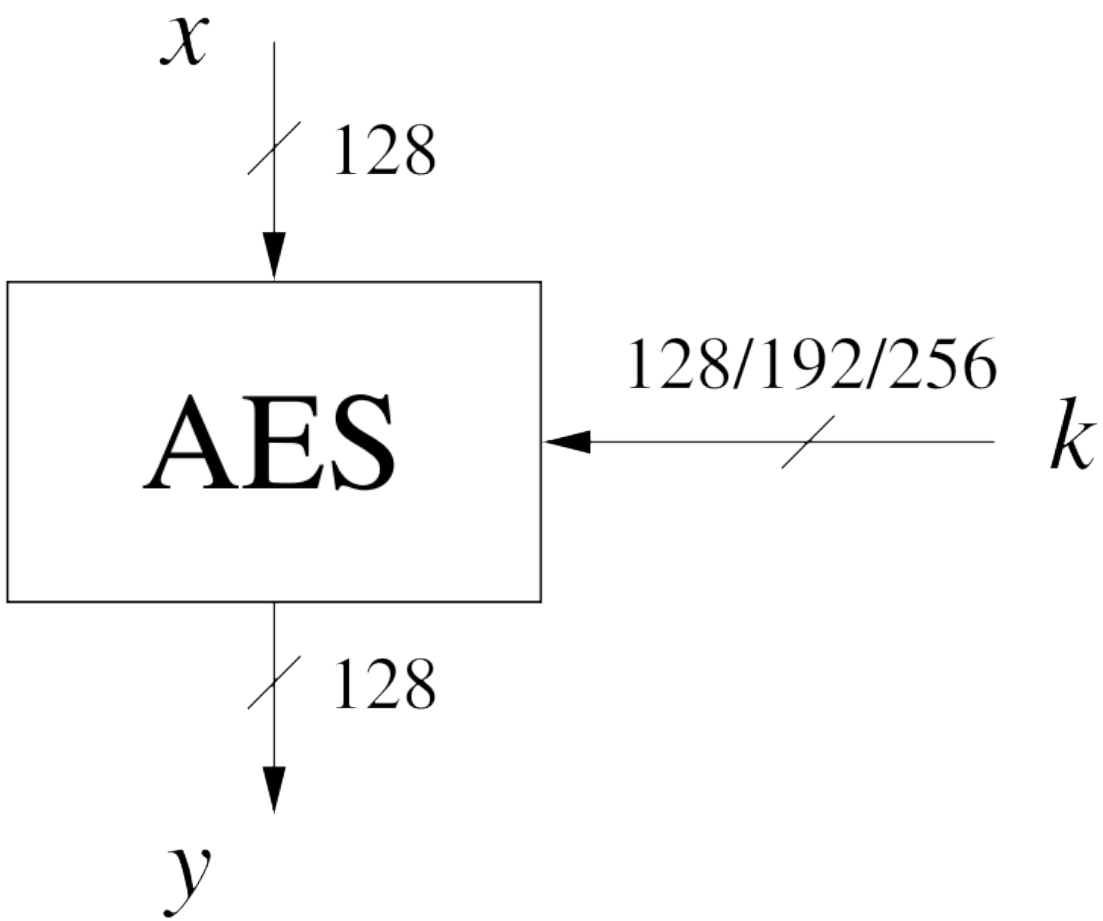
Finalists: MARS, RC6, Rijndael, Serpent, Twofish

Security policy note

In 2003, NSA allowed AES for **SECRET** data with all key sizes and for **TOP SECRET** data with 192/256-bit keys.

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Advanced Encryption Standard (AES)



Rijndael can use other block sizes (128,160,192,224,or 256 bits);
AES uses only 128 bits.
Finite-field arithmetic is fundamental to AES.

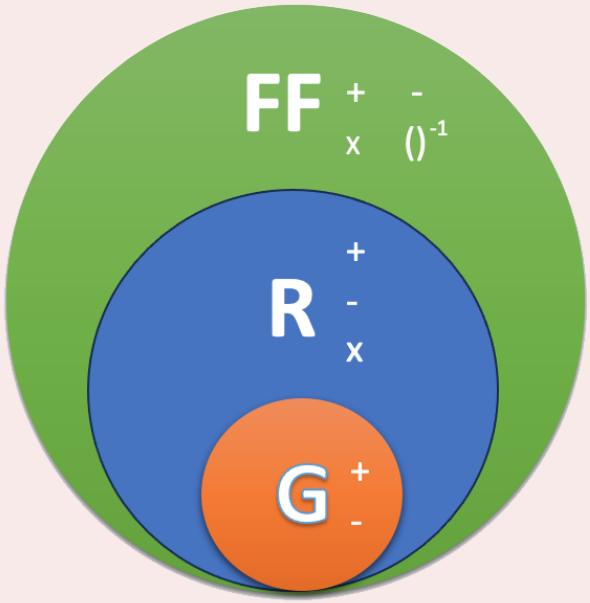
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Galois Fields

Terminology

Basic Algebraic Structures

- Galois field = Finite Field
- Group, Ring, Finite Field



Group as an Algebraic Structure

Definition

A group is an algebraic structure formed by a set G and one operation $\circ$:

$$(G, \circ)$$

The operation must satisfy four properties.

1. **Closure**

$$a, b \in G \Rightarrow a \circ b \in G$$

2. **Associativity**

$$(a \circ b) \circ c = a \circ (b \circ c)$$

3. **Identity**

$$a \circ e = e \circ a = a$$

4. **Inverse**

$$a \circ a^{-1} = a^{-1} \circ a = e$$

Example: $(\mathbb{Z}_5, +)$

$$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$$

Addition is performed modulo 5:

$$3 + 4 = 7 \equiv 2 \pmod 5$$

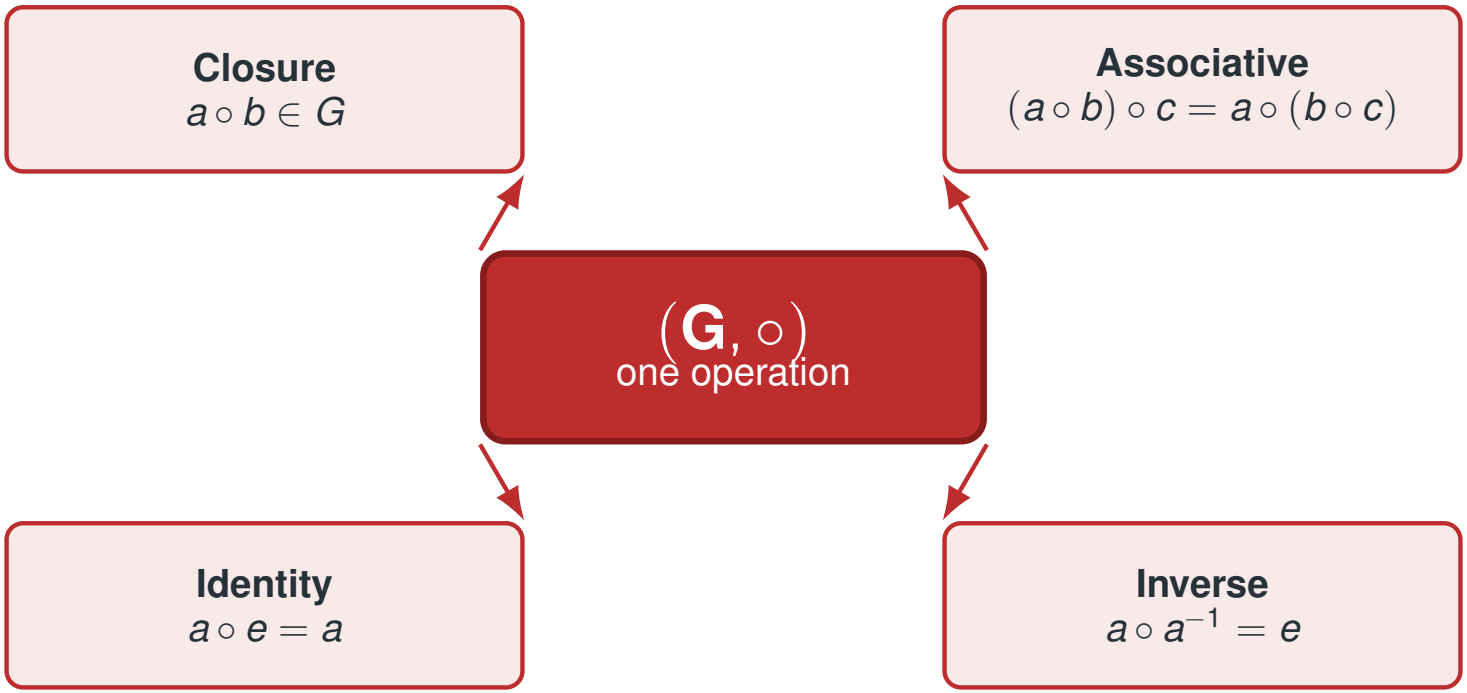
Identity:

$$e = 0$$

Some inverses:

$$1 + 4 \equiv 0 \pmod 5$$

Group as an Algebraic Structure



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Abelian Groups

Definition

An Abelian group is a group $(G, \circ)$ in which the operation is also **commutative**:

$$a \circ b = b \circ a, \qquad \forall\, a, b \in G$$

Therefore, an Abelian group satisfies the five properties below.

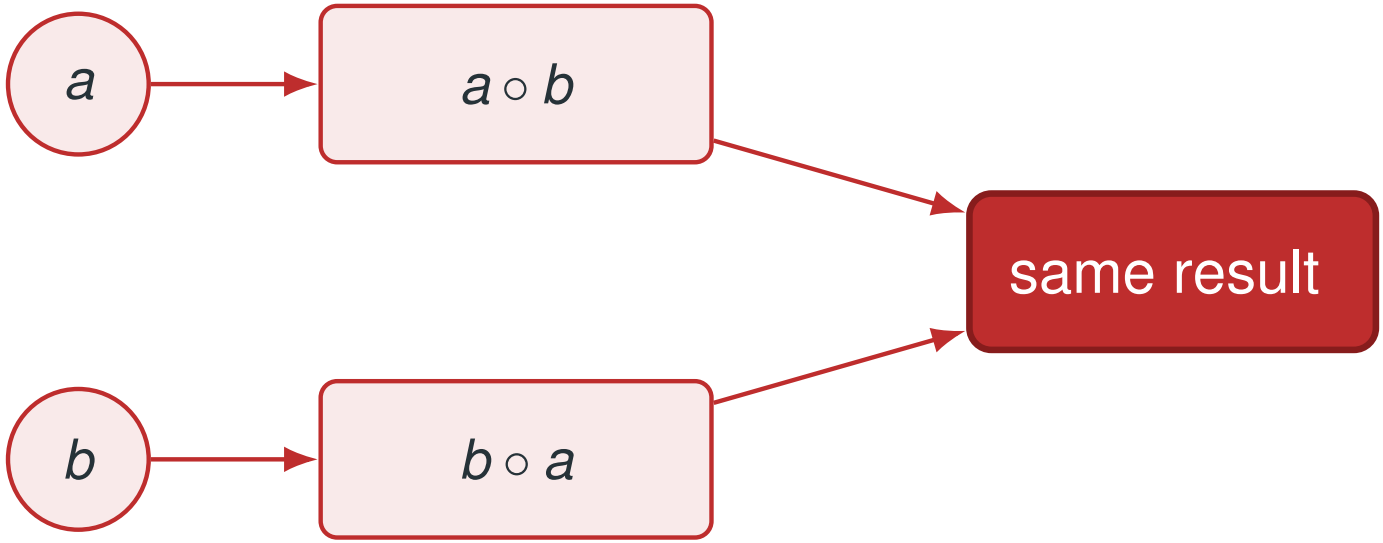
1. **Closure**
$$a, b \in G \Rightarrow a \circ b \in G$$
2. **Associativity**
$$(a \circ b) \circ c = a \circ (b \circ c)$$
3. **Identity**
$$a \circ e = e \circ a = a$$
4. **Inverse**
$$a \circ a^{-1} = a^{-1} \circ a = e$$
5. **Commutativity**
$$a \circ b = b \circ a$$

Every Abelian group is a group, but **not every group is Abelian**.

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Abelian Group: Order Does Not Matter

$$a \circ b = b \circ a$$



Example in $\mathbb{Z}_5$

$$2 + 4 \equiv 4 + 2 \equiv 1 \pmod 5$$

Abelian Group = Group + Commutative Operation

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Rings

Definition

A ring is an algebraic structure with a set R and two operations:

$$(R, +, \cdot)$$

The two operations are **addition** and **multiplication**.

Main Properties:

- $(R, +)$ is an **Abelian group**.
- Multiplication is **closed**:

$$a, b \in R \Rightarrow a \cdot b \in R$$

- Multiplication is **associative**:

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

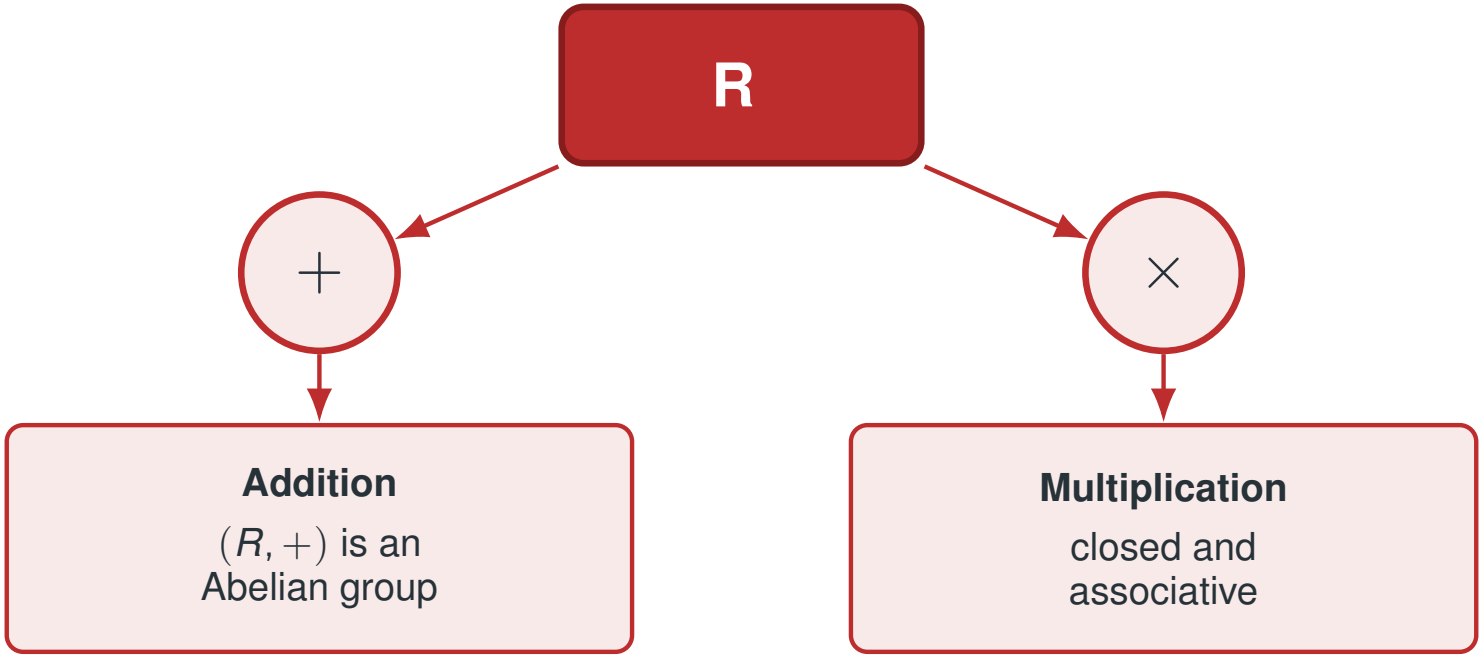
- Multiplication is distributive over addition:

$$a(b + c) = ab + ac$$

$$(a + b)c = ac + bc$$

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Rings



Connection Between Both Operations

$$a(b+c) = ab + ac \qquad (a+b)c = ac + bc$$

Multiplicative inverses are not required.

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Finite Fields

Definition

A finite field is a field with a **finite number of elements**. It is usually written as $\mathrm{GF}(q)$ where

$$q = p^m$$

p is a prime number and m is a positive integer.

Main Properties

A finite field has two operations: **addition** and **multiplication**.

- $(F, +)$ is an **Abelian group**.
- Non-zero elements form an Abelian group under multiplication.
- Every non-zero element has a **multiplicative inverse**:

$$a \cdot a^{-1} = 1$$

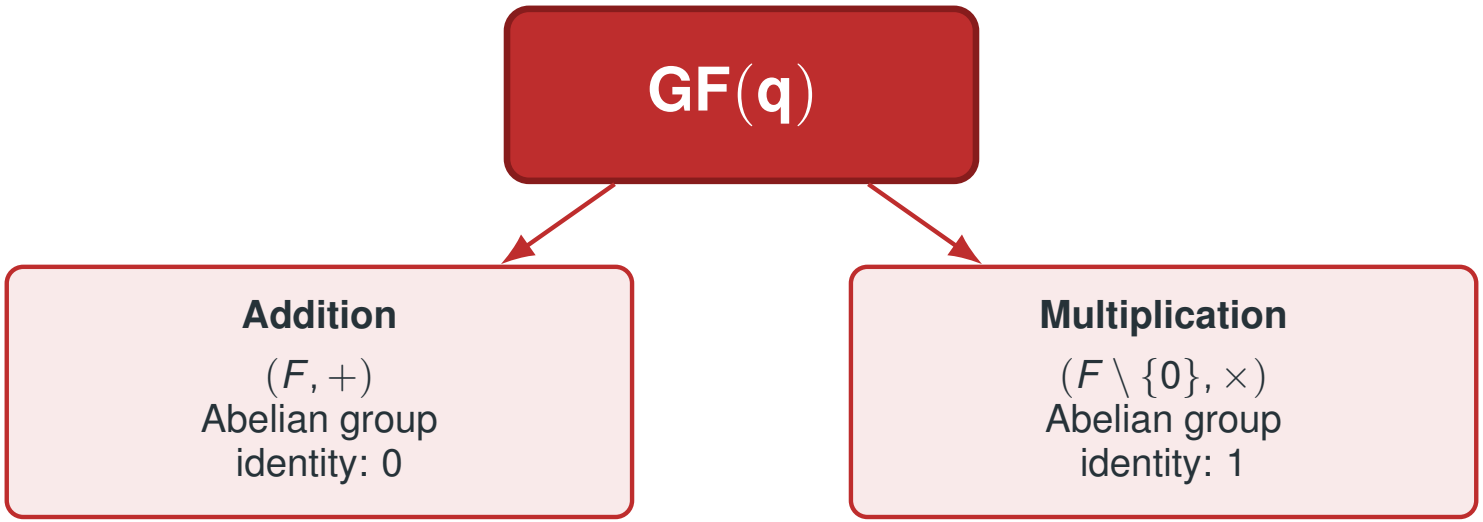
- Multiplication is distributive over addition:

$$a(b + c) = ab + ac$$

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Example: GF(5)

Finite Field



Main Property

$$a \neq 0 \implies \exists\, a^{-1} \quad \text{such that} \quad a\,a^{-1} = 1$$

We can divide by every nonzero element.

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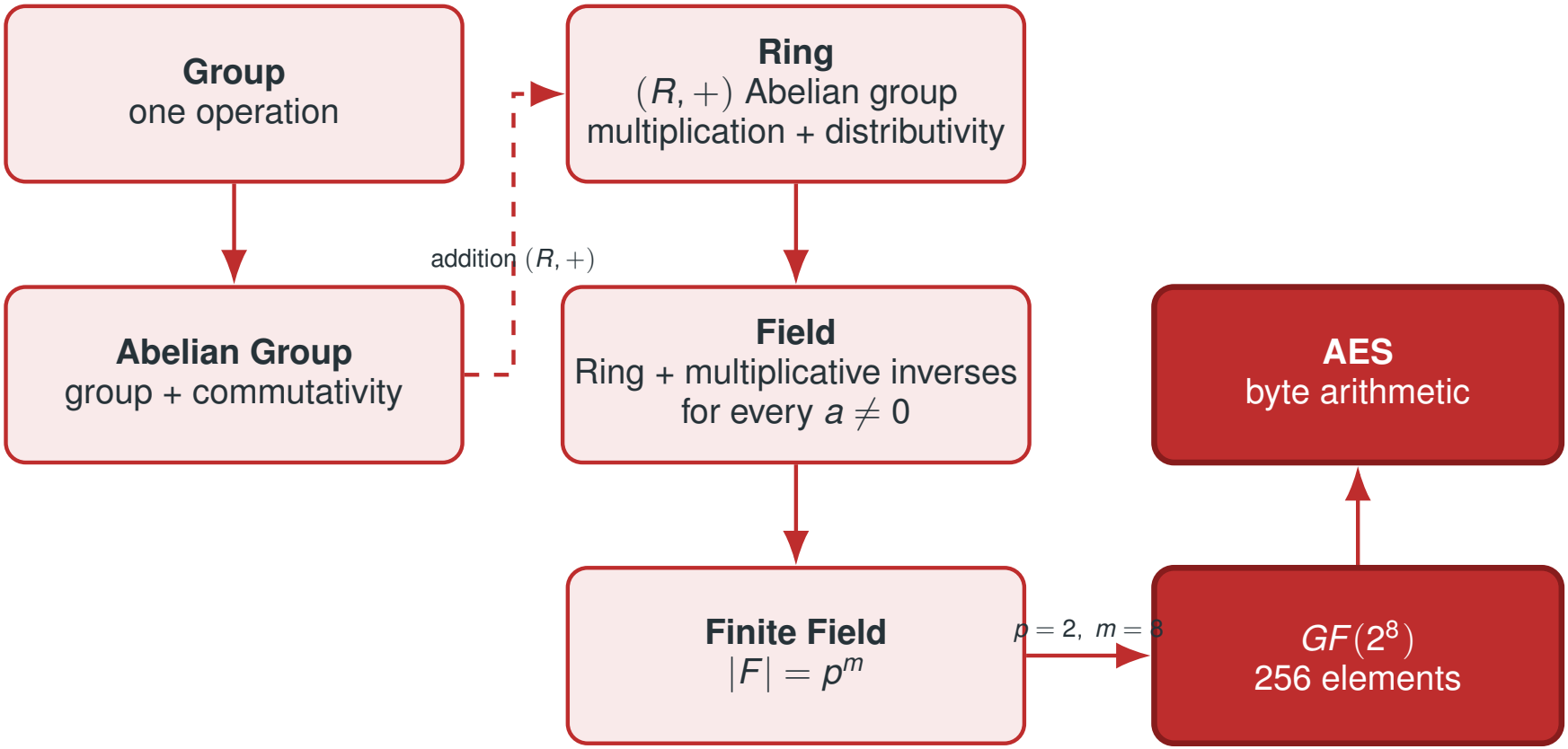
Finite Field

Formal definition of fields

- A finite field, sometimes also called **Galois field** *GF*, is a set with a finite number of elements.
- A Finite Field is a set of numbers in which we can **add, substract, multiply and divide**.
- **In Cryptography, we need Finite Fields**.

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From Algebraic Structures to AES



AES uses the finite field $GF(2^8)$ for byte arithmetic.

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When Does a Finite Field Exist?

$$\mathbf{q = p^m}$$

p is prime
$2, 3, 5, 7, 11, \ldots$

m is positive
$1, 2, 3, \ldots$

Finite Field

$$GF(q) \text{ exists} \iff q = p^m$$

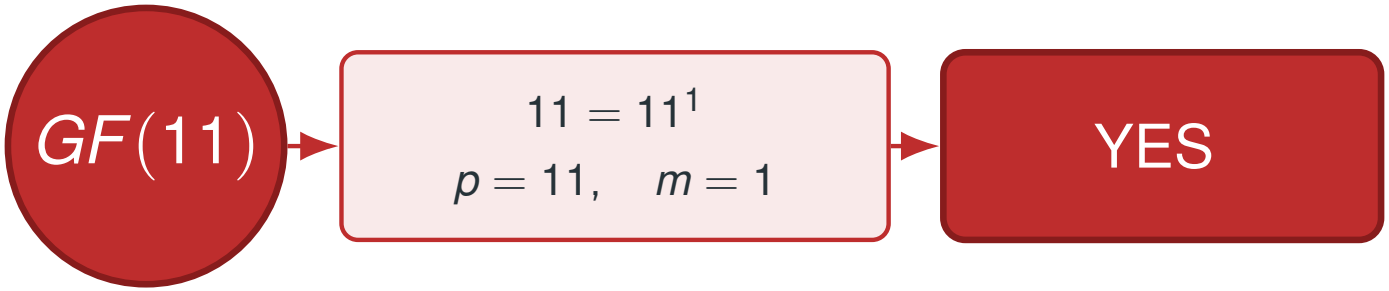
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Example 1: Does $GF(11)$ Exist?



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Example 1: Does $GF(11)$ Exist?



$GF(11)$ exists

11 is a **prime number**.

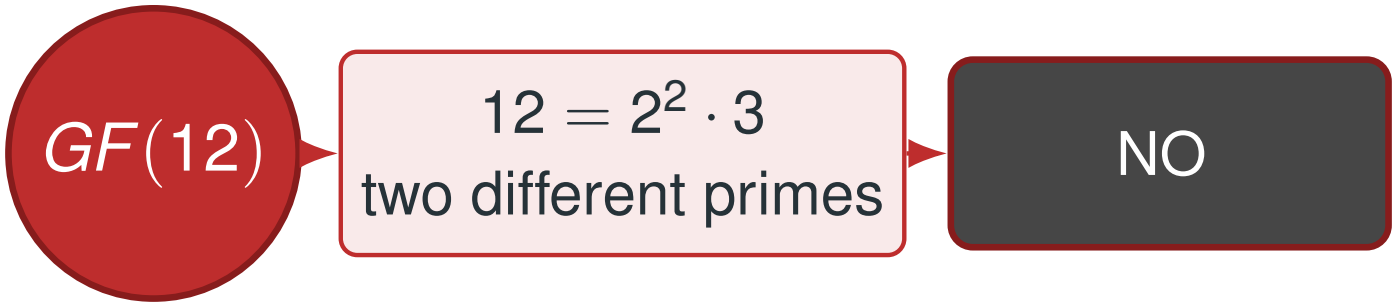
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Example 2: Does $GF(12)$ Exist?



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Example 2: Does $GF(12)$ Exist?



$$12 \neq p^m$$

$GF(12)$ does not exist

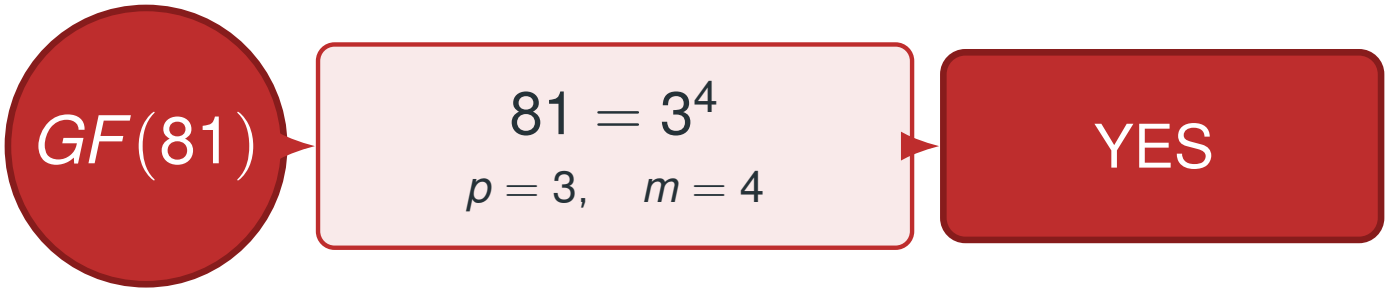
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Example 3: Does $GF(81)$ Exist?



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Example 3: Does $GF(81)$ Exist?

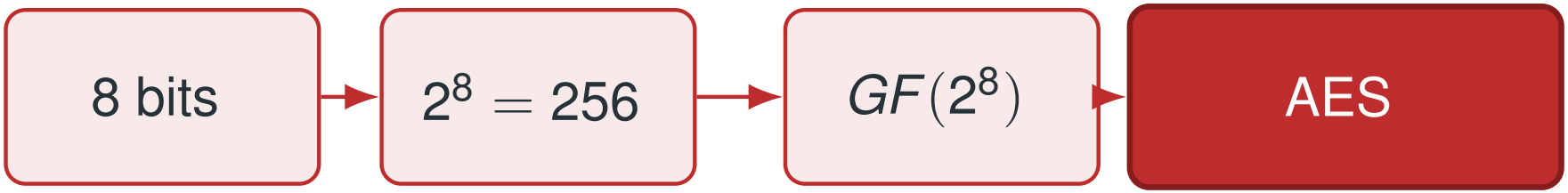


$$GF(81) = GF(3^4)$$

3 is prime and 81 is a **prime power**.

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Example 4: Why Does AES Use $GF(2^8)$?



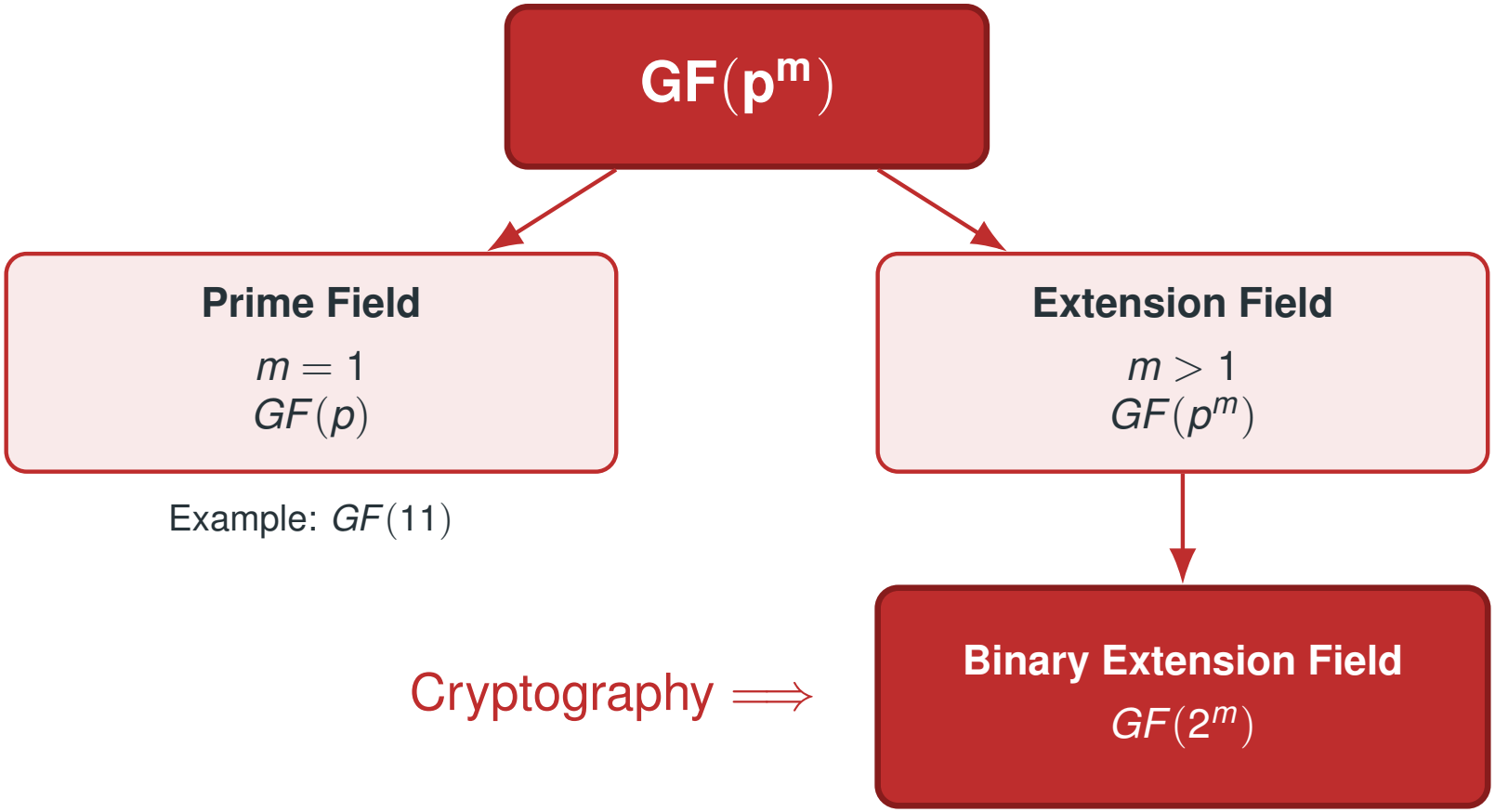
$$\boxed{256 = 2^8} \qquad p = 2, \quad m = 8$$

AES Connection

$$\boxed{\text{1 byte}} \iff \boxed{\text{1 element of } GF(2^8)}$$

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Types of Finite Fields (cont.)



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Types of Finite Fields (cont.)

AES

$$\boxed{GF(2^8) = GF(256)} \quad \Longrightarrow \quad \boxed{\text{1 element = 1 byte}}$$

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Prime Field Arithmetic

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Prime Field Arithmetic (cont.)

Prime Field

$$\boxed{GF(p) = \{0, 1, 2, \ldots, p-1\}} \qquad p \text{ is prime}$$

Addition

$$a + b \pmod{p}$$

Subtraction

$$a - b \pmod{p}$$

Multiplication

$$a \cdot b \pmod{p}$$

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Prime Field Arithmetic (cont.)

Example: $GF(7) = \{0, 1, 2, 3, 4, 5, 6\}$

$$5 + 4 \equiv 2 \pmod 7 \qquad 5 - 6 \equiv 6 \pmod 7 \qquad 5 \cdot 4 \equiv 6 \pmod 7$$

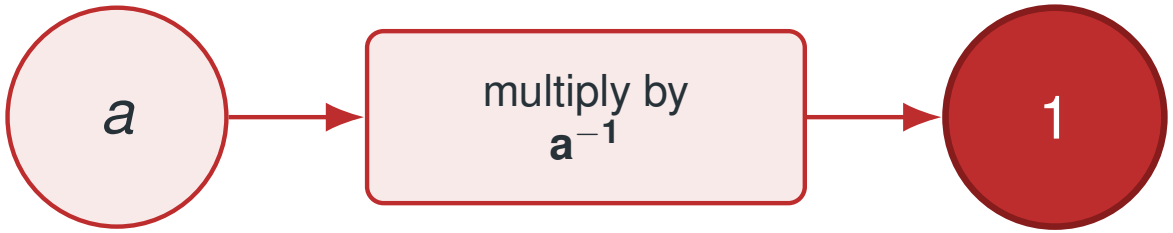
The result always remains inside $GF(p)$.

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Multiplicative Inverse in $GF(p)$ (cont.)

Prime Field

$$GF(p) = \{0, 1, 2, \ldots, p-1\}, \qquad p \text{ is prime}$$



$$\boxed{a \cdot a^{-1} \equiv 1 \pmod{p}} \qquad a \neq 0$$

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Multiplicative Inverse in $GF(p)$ (cont.)

Example: inverse of 3 in $GF(7)$

$$3 \cdot 5 = 15 \equiv 1 \pmod 7$$

$$\boxed{3^{-1} = 5 \quad \text{in } GF(7)}$$

Zero has no multiplicative inverse.

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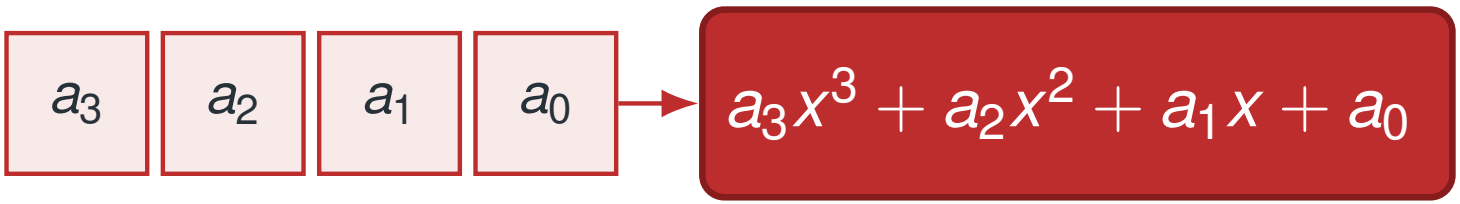
Extension Field $GF(2^m)$ Arithmetic

Element Representation

The elements of $GF(2^m)$ are polynomials:

$$A(x) = a_{m-1}x^{m-1} + \cdots + a_1 x + a_0$$

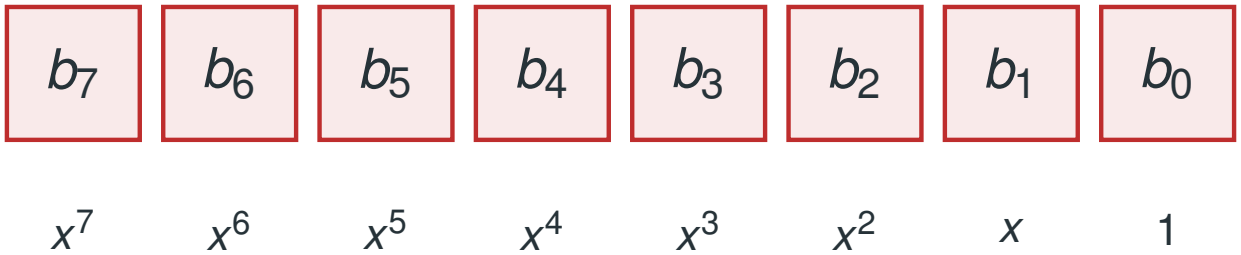
$$a_i \in GF(2) = \{0, 1\}, A(x) \in GF(2^m)$$



Each coefficient is only 0 or 1.

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Bytes as Elements of $GF(2^8)$ (cont.)



$$b_7x^7 + b_6x^6 + b_5x^5 + b_4x^4 + b_3x^3 + b_2x^2 + b_1x + b_0$$

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Bytes as Elements of $GF(2^8)$ (cont.)

Example: $\mathtt{B9}_{16}$

$$\mathtt{B9}_{16} = 10111001_2$$

$$1x^7 + 0x^6 + 1x^5 + 1x^4 + 1x^3 + 0x^2 + 0x + 1$$

$$\boxed{x^7 + x^5 + x^4 + x^3 + 1}$$

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Example: Elements of $GF(2^3)$

$$A(x) = a_2x^2 + a_1x + a_0$$

Binary Representation	Polynomial Representation
000	0
001	1
010	x
011	$x + 1$
100	x^2
101	$x^2 + 1$
110	$x^2 + x$
111	$x^2 + x + 1$

How do we add and multiply these elements?

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Addition and Subtraction in $GF(2^m)$

Main Rule

Polynomial coefficients are computed in $GF(2) = \{0, 1\}$.

$$0 + 0 = 0 \qquad 0 + 1 = 1 \qquad 1 + 0 = 1 \qquad 1 + 1 = 0$$

$$1 + 1 = 0 \quad \text{and} \quad 1 - 1 = 0$$

$$\boxed{\text{Addition} = \text{Subtraction} = \text{XOR}}$$

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Example: Addition in $GF(2^3)$

Polynomial View

$$A(x) = x^2 + x + 1$$
$$B(x) = x^2 + 1$$
$$A(x) + B(x) = (1+1)x^2 + x + (1+1)$$
$$= 0x^2 + x + 0$$
$$= \boxed{x}$$

Binary View

$$\begin{array}{r} 111 \\ \oplus\ 101 \\ \hline 010 \end{array}$$

$$\boxed{010 \longleftrightarrow x}$$

Same operation, two representations

$$\boxed{A(x) + B(x) = A(x) - B(x)}$$

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Exercise: Addition in $GF(2^8)$

Calculate:

$$\boxed{\mathrm{E7}_{16} + \mathrm{8A}_{16}}$$

using polynomial representation in $GF(2^8)$.

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Solution: Addition in $GF(2^8)$

Polynomial Addition

$$A(x) = x^7 + x^6 + x^5 + x^2 + x + 1$$

$$B(x) = x^7 + x^3 + x$$

$$\begin{aligned} A(x) + B(x) &= (x^7 + x^6 + x^5 + x^2 + x + 1) \\ &\quad + (x^7 + x^3 + x) \\ &= 2x^7 + x^6 + x^5 + x^3 + x^2 + 2x + 1 \end{aligned}$$

$$A(x) + B(x) = \boxed{x^6 + x^5 + x^3 + x^2 + 1}$$

$$= 01101101_2 = \boxed{6D_{16}}$$

Exercise: Addition Using XOR

Calculate

$$\boxed{\mathrm{E7}_{16} \oplus \mathrm{8A}_{16}}$$

using binary representation.

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Solution: Addition Using XOR

Bitwise XOR

$$\begin{array}{r} 11100111 \\ \oplus\ 10001010 \\ \hline 01101101 \end{array}$$



Addition in $GF(2^8)$ is equivalent to XOR.

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Multiplication in $GF(2^m)$ (cont.)

Main Rule

Multiply the polynomials normally, but compute all coefficients in

$$\boxed{GF(2) = \{0, 1\}}$$

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Multiplication in $GF(2^m)$ (cont.)

Example in $GF(2^3)$

$$A(x) = x^2 + x + 1$$
$$B(x) = x^2 + 1$$

$$\begin{aligned} A(x)B(x) &= (x^2 + x + 1)(x^2 + 1) \\ &= x^4 + x^3 + x^2 + x^2 + x + 1 \\ &= x^4 + x^3 + \underbrace{(1+1)}_{0} x^2 + x + 1 \\ &= \boxed{x^4 + x^3 + x + 1} \end{aligned}$$

Multiplication in $GF(2^m)$ (cont.)

But this result is not yet an element of $GF(2^3)$.

$$\deg\left(x^4 + x^3 + x + 1\right) = 4 \qquad > \qquad m - 1 = 2$$

Next step: polynomial reduction

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Polynomial Reduction in $GF(2^3)$ (cont.)

Why Do We Need an Irreducible Polynomial?

The product must be reduced to remain inside the field.

$$C(x) = x^4 + x^3 + x + 1 \xrightarrow{\text{reduce modulo}} P(x) = x^3 + x + 1$$

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Polynomial Reduction in $GF(2^3)$ (cont.)

Irreducible Polynomial for this Example

$$\boxed{P(x) = x^3 + x + 1}$$

$$C(x) \bmod P(x)$$

An irreducible polynomial plays a role similar to a prime number.

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Example: Reduction in $GF(2^3)$ (cont.)

Given

$$C(x) = x^4 + x^3 + x + 1$$

$$P(x) = x^3 + x + 1$$

Reduction

$$C(x) = x^4 + x^3 + x + 1$$

$$C(x) + xP(x) = x^3 + x^2 + 1$$

$$C(x) + xP(x) + P(x) = x^2 + x$$

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Example: Reduction in $GF(2^3)$ (cont.)

$$\boxed{C(x) \bmod P(x) = x^2 + x}$$

In $GF(2)$: subtraction = addition = XOR

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Why Do We Need an Irreducible Polynomial? (cont.)

Prime Field

$GF(p)$

$\boxed{\text{mod } p}$

Example:

$GF(7)$

Extension Field

$GF(2^m)$

$\boxed{\text{mod } P(x)}$

$P(x)$ irreducible

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Why Do We Need an Irreducible Polynomial? (cont.)

Irreducible polynomial
$\approx$ prime number

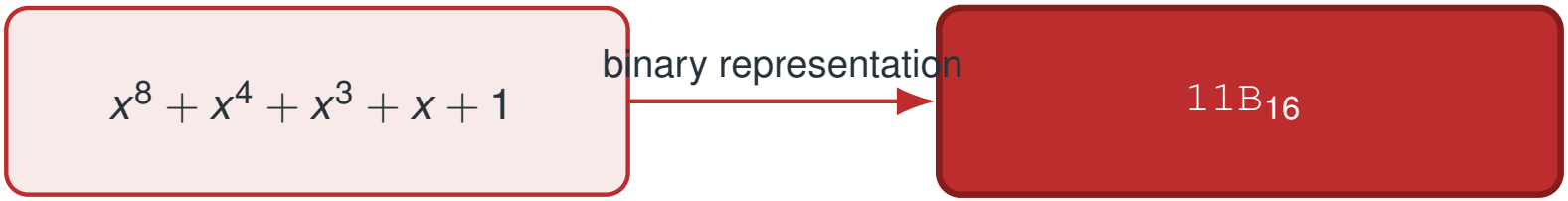
Several irreducible polynomials of degree m may exist.

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The AES Irreducible Polynomial (cont.)

AES Works in $GF(2^8)$

$$P(x) = x^8 + x^4 + x^3 + x + 1$$



$$P(x) \equiv 0$$

therefore

$$x^8 \equiv x^4 + x^3 + x + 1 \pmod{P(x)}$$

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The AES Irreducible Polynomial (cont.)

This relation reduces terms of degree 8 or higher.

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Exercise: Multiplication in $GF(2^8)$ (cont.)

Problem

Calculate

$$\boxed{02_{16} \cdot 87_{16}}$$

in AES $GF(2^8)$.

AES Reduction Polynomial

$$P(x) = x^8 + x^4 + x^3 + x + 1$$

What is the result?

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Solution: Multiplication in $GF(2^8)$ (cont.)

Polynomial Multiplication

$$x(x^7 + x^2 + x + 1) = x^8 + x^3 + x^2 + x$$

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Solution: Multiplication in $GF(2^8)$ (cont.)

Reduce x^8

Using

$$x^8 \equiv x^4 + x^3 + x + 1$$

we obtain

$$\begin{aligned} x^8 + x^3 + x^2 + x &\equiv (x^4 + x^3 + x + 1) + x^3 + x^2 + x \\ &= x^4 + (1+1)x^3 + x^2 + (1+1)x + 1 \\ &= \boxed{x^4 + x^2 + 1} \end{aligned}$$

$$x^4 + x^2 + 1 \longleftrightarrow 00010101_2 \longleftrightarrow \boxed{15_{16}}$$

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Extension Field GF(2^m) Arithmetic

GF(2^3)

$$C(x) = x^4 + x^3 + x + 1$$

$$P(x) = x^3 + x + 1$$

C(x) mod P(x)

$$\begin{array}{rr|l} x^4 + x^3 + x + 1 & & x^3 + x + 1 \\ \cline{3-3} -x^4 - x^2 - x & & x + 1 \\ \cline{1-1} x^3 - x^2 + 1 & & \\ -x^3 - x - 1 & & \\ \cline{1-1} -x^2 - x & & \end{array}$$

$$C(x) \mod P(x) = x^2 + x$$

For every field GF(2^m) there are several irreductible polynomial

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Extension Field GF(2^m) Arithmetic

The AES irreductible polynomial

$$P(x) = x^8 + x^4 + x^3 + x + 1$$

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Extension Field GF(2^m) Arithmetic

Polynomial Reduction

- To ensure that all calculations stay within the finite field, it's necessary to reduce polynomials of higher degree.
- Polynomials of degree 8 or more can always be reduced, using the fact that in the finite field, we have $f(x) = 0$, so we have the relation: $x^8 + x^4 + x^3 + x + 1 = 0$
  - $x^8 = x^4 + x^3 + x + 1$



Type some Sage code below and press Evaluate.

```
1 F2 = GF(2)
2 K.<x> = F2[]
3 f = x^8 + x^4 + x^3 + x + 1
4 f.is_irreducible()
```

Evaluate

```
True
```

Extension Field GF(2^m) Arithmetic

Polynomial Reduction

- To carry out a product of any two polynomials then, use the relation $x^8 = x^4 + x^3 + x + 1$ to eliminate any terms $k \geq 8$ reducing modulo 2.

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Extension Field GF(2^m) Arithmetic

Polynomial Reduction

- 00000010_2 corresponds to the polynomial $0x^7 + 0x^6 + 0x^5 + 0x^4 + 0x^3 + 0x^2 + 1x^1 + 0$.
- 10000111_2 corresponds to $x^7 + x^2 + x + 1$. So to do the multiplication, we calculate:

$$\begin{aligned} x \cdot (x^7 + x^2 + x + 1) &= x^8 + x^3 + x^2 + x \ mod\ 2 \\ &= (x^4 + x^3 + x + 1) + x^3 + x^2 + x \ mod\ 2 \\ &= (x^4 + 2x^3 + x^2 + 2x + 1 \ mod\ 2 \\ &= x^4 + x^2 + 1 \ mod\ 2 \end{aligned}$$

- $00010101_2 = 15_{HEX}$

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Inversion in $GF(2^m)$ (cont.)

Definition

For a nonzero element

$$A(x) \in GF(2^m), \qquad A(x) \neq 0,$$

its multiplicative inverse $A^{-1}(x)$ satisfies

$$\boxed{A(x) \cdot A^{-1}(x) \equiv 1 \pmod{P(x)}}$$

Key Idea

$\boxed{A^{-1}(x) \text{ is the element that “undoes” } A(x)}$

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Inversion in $GF(2^m)$ (cont.)

Zero has no multiplicative inverse.

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Extension Field GF(2^m) Arithmetic

Inversion in GF(2^8)

- Inversion in GF(2^8) is the core operation of the Byte Substitution transformation, which contains the AES S-Boxes.
- For a given finite field GF(2^8) and the corresponding irreducible reduction polynomial P(x), the inverse A^{-1} of a nonzero element $A \in$ GF(2^8) is defined as:

$$A^{-1}(x) \cdot A(x) = 1 \ mod \ P(x) \tag{1}$$

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Extension Field GF(2^m) Arithmetic

Multiplicative inverse table in $GF(2^8)$ for bytes xy used within the AES S-Box

X \ Y	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	00	01	8D	F6	CB	52	7B	D1	E8	4F	29	C0	B0	E1	E5	C7
1	74	B4	AA	4B	99	2B	60	5F	58	3F	FD	CC	FF	40	EE	B2
2	3A	6E	5A	F1	55	4D	A8	C9	C1	0A	98	15	30	44	A2	C2
3	2C	45	92	6C	F3	39	66	42	F2	35	20	6F	77	BB	59	19
4	1D	FE	37	67	2D	31	F5	69	A7	64	AB	13	54	25	E9	09
5	ED	5C	05	CA	4C	24	87	BF	18	3E	22	F0	51	EC	61	17
6	16	5E	AF	D3	49	A6	36	43	F4	47	91	DF	33	93	21	3B
7	79	B7	97	85	10	B5	BA	3C	B6	70	D0	06	A1	FA	81	82
8	83	7E	7F	80	96	73	BE	56	9B	9E	95	D9	F7	02	B9	A4
9	DE	6A	32	6D	D8	8A	84	72	2A	14	9F	88	F9	DC	89	9A
A	FB	7C	2E	C3	8F	B8	65	48	26	C8	12	4A	CE	E7	D2	62
B	0C	E0	1F	EF	11	75	78	71	A5	8E	76	3D	BD	BC	86	57
C	0B	28	2F	A3	DA	D4	E4	0F	A9	27	53	04	1B	FC	AC	E6
D	7A	07	AE	63	C5	DB	E2	EA	94	8B	C4	D5	9D	F8	90	6B
E	B1	0D	D6	EB	C6	0E	CF	AD	08	4E	D7	E3	5D	50	1E	B3
F	5B	23	38	34	68	46	03	8C	DD	9C	7D	A0	CD	1A	41	1C

Inversion in GF(2^8)

- Table shows the values which are used within the S-Box of AES.

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Extension Field GF(2^m) Arithmetic

Inversion in GF(2^8)

From Table the inverse of:

$$x^7 + x^6 + x = (11000010)_2 = (C2)_{hex} = (xy)$$

is given by the element in row C, column 2: $(2F)_{hex} = (00101111)_2 = x^5 + x^3 + x^2 + x + 1$

This can be verified by multiplication:

$$(x^7 + x^6 + x)\cdot(x^5 + x^3 + x^2 + x + 1) = 1 \bmod P(x).$$

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Example: Inverse and AES S-box

Multiplicative Inverse

$$\mathtt{C2}_{16} = 11000010_2 \longleftrightarrow x^7 + x^6 + x$$

$$\boxed{\mathtt{C2}_{16}^{-1} = \mathtt{2F}_{16}}$$

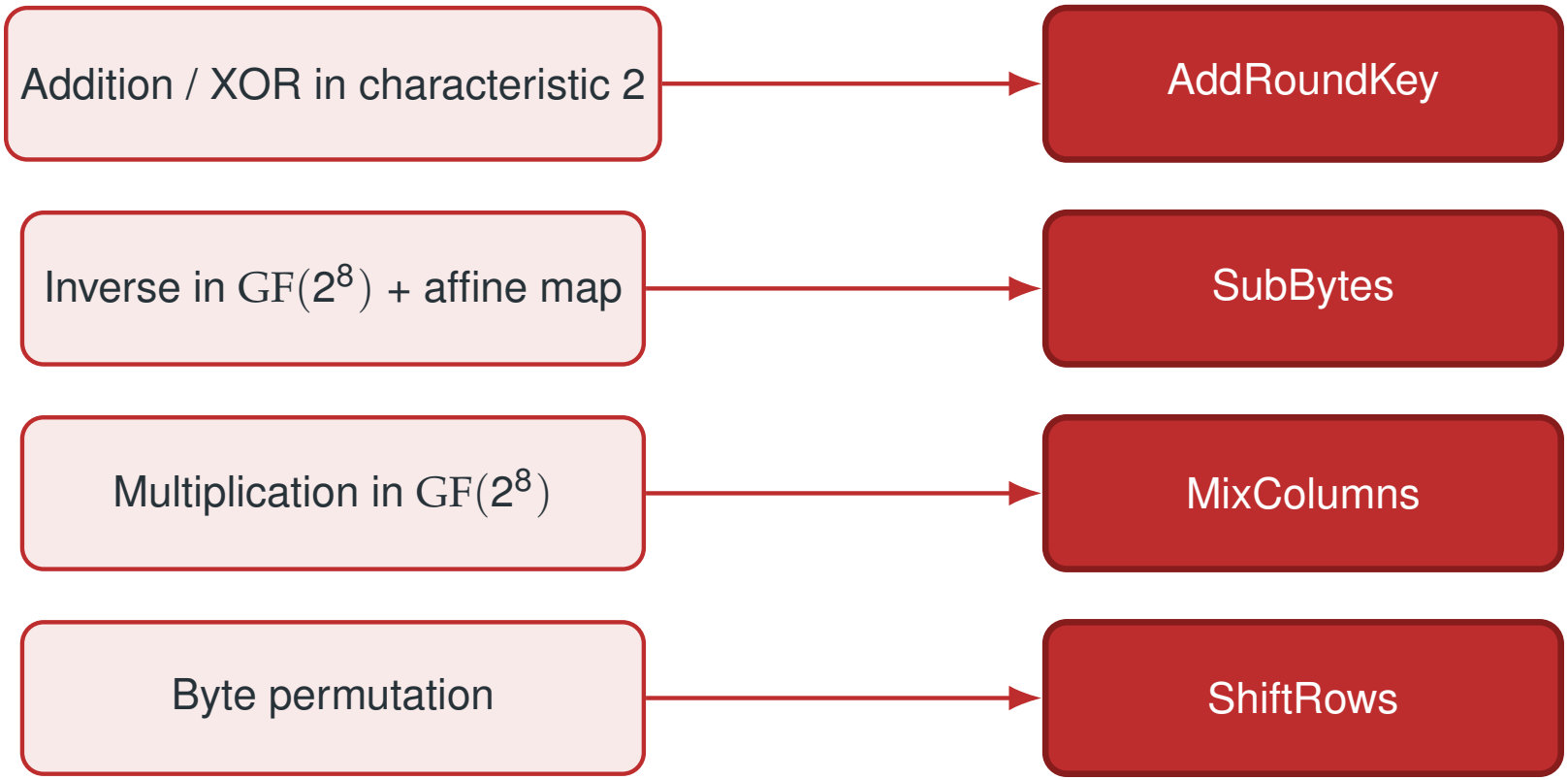
$$\mathtt{2F}_{16} = 00101111_2 \longleftrightarrow x^5 + x^3 + x^2 + x + 1$$

$$\mathtt{C2}_{16} \xrightarrow{\text{inverse in } \mathrm{GF}(2^8)} \mathtt{2F}_{16} \xrightarrow{\text{affine map}} \boxed{25_{16}}$$

$$S(\mathtt{C2}_{16}) = \mathtt{25}_{16}$$

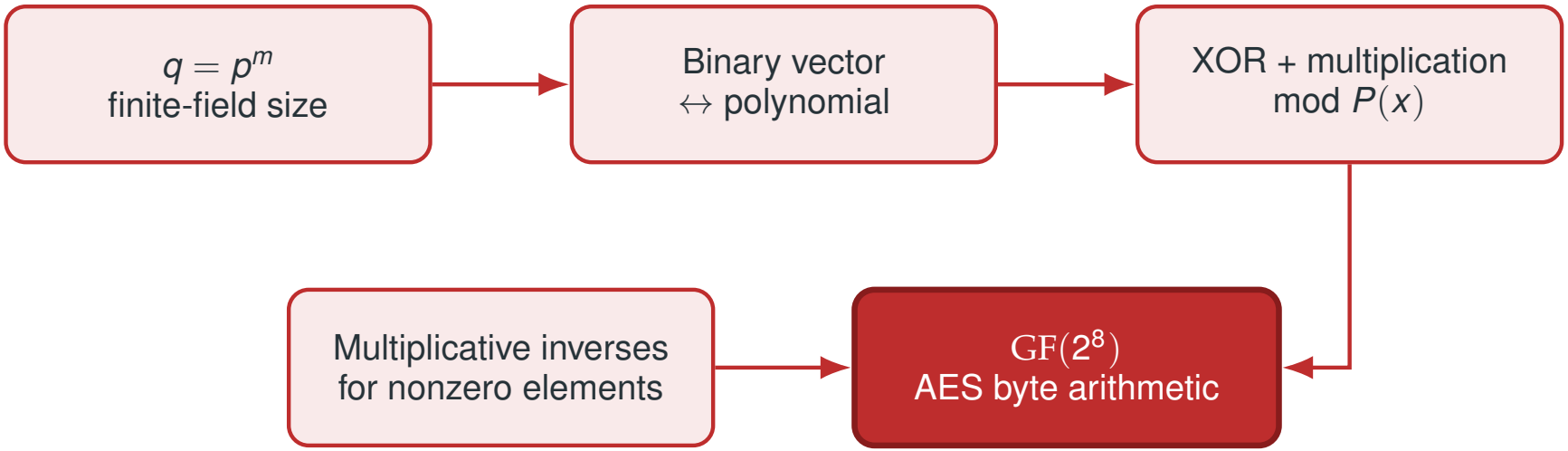
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Why Finite Fields Matter in AES



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Key Takeaways



Finite fields give AES efficient, invertible, byte-oriented arithmetic.

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Cryptography

AES & Galois Fields

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